Moving-meridian source estimates for strict Conway tangles

Abstract

We estimate the actual Newton contraction of a fixed ordered Conway tangle when its main meridian moves on the scale of the inverse level. At a product junction the deformed trace need not be bounded by its complete value at the same indices. We prove a reflected-index comparison instead, and assemble it over arbitrary connected regions of product junctions. The resulting error has a strict exponential loss at the original shared pressure. The retained finite sum has primitive wire coordinates, a nonzero holomorphic saddle and a shared angular bound. Its actual crossing traces have one nonzero joint first-charge contribution for independently long fixed signed powers. The proof includes both level parities, all first normal coordinates, and fixed translated nonreal meridian charts. It supplies the analytic source needed for higher meridian expansions; no longitude identification is used.

1 The actual source and the assertions

Fix a literal ordered expression in products, physical quarter-turns and crossing boxes. Consecutive products without a turn are combined into one junction. A junction carrying an original crossing operator is a *circle vertex*; a junction with no such operator is *bare*. Every bare junction has valence at least three, counting its external attachments. The reduced graph is a tree. Its endpoint orders, orientations, closing cut and framing remain fixed. Identity coupons at bare vertices are not additional crossing circles.

We require the expression to be strict in the sense of [3, Lemma 2.2 and Theorem 2.5]. Explicitly its positive Newton state has leaf labels in $(0, 1/2)$ and, after deleting only the forced final flat gate, each multiplication has

$$\max(x, y) < z < x + y < 1.$$

All hidden indices in an ordered binary resolution are retained. Theorem 3.3 of [3] proves that the resulting compact positive action has a unique strict maximum, with negative definite Hessian. Write its value as F_0. If a knot invariant is mentioned, the original endpoint tracing must give one knot, with no component closed inside a proper tangle. There are $H \geq 2$ original circle vertices. The diagram and its integer powers are fixed before the level tends to infinity.

Here are the exact finite conventions. For $N \geq 2$ put

$$\begin{gathered} q = e^{\pi \mathrm{i}/N}, \quad d = N - 1, \quad m = \lfloor d/2 \rfloor, \quad S_r = \prod_{a=1}^{r} 2\sin(\pi a/N), \\ c_j = 2\cos((2j+1)\pi/N), \qquad e_r(z) = S_r^{-2} \prod_{j<r} (c_j - z). \end{gathered} \tag{1.1}$$

The empty product is one. A low index satisfies $r \leq m$; a high index satisfies $r > m$. These are basis indices, not ordinary circle colors. Let $\mathsf{T}_N(u + u^{-1}) = u^N + u^{-N}$ and define

$$P_\sigma(z) = \mathsf{T}_N(z) + 2\cos(2\pi\sigma), \qquad \phi_\sigma(F) = (-1)^d N^2 [z^d] \operatorname{rem}_{P_\sigma} F. \tag{1.2}$$

The parameter of an actual vertex is $\sigma = s, -s$, or 0, according to its ordered orientation type. Moving equal-color vertices are nonadjacent across physical turns; constant vertices may be adjacent. This follows from the six domain/codomain types in [1, Theorem 1.1], not from an assumed bipartite coloring.

The same finite theorem supplies the full pairing $\phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}$. After lowering its two indices, an original physical turn edge has weight

$$\mathbf{1}_{i=j}\gamma_{e,N}(s)\frac{H_N(s)}{W_i(s)}, \qquad H_N(s) = \frac{\sin \pi s}{N \sin(\pi s/N)}, \qquad W_i(s) = \prod_{a=1}^{i}\left(1 - \frac{\sin^2(\pi s/N)}{\sin^2(\pi a/N)}\right). \tag{1.3}$$

Here $\gamma_{e,N} \in \{1, q^{s^2/2}, q^{-s^2/2}\}$ is the prescribed typed scalar. The quotient and its carriers at each vertex are those of its actual ordered object. Theorem 7.1 of [1] gives the finite contraction with all $0 \le i < N$, and exactly one outside factor H_N for the reduced cut. It allows identity coupons, arbitrary valence and any summation root.

The elementary action functions used throughout are

$$J(x) = \int_0^x \log(2 \sin \pi t)\, dt, \qquad g(x) = J(2x) - 4J(x), \quad 0 \le x \le \tfrac{1}{2}, \tag{1.4}$$

$$\begin{aligned} m(x,y,z) = {} & J(x+y) + J(z) - J(x) - J(y) \\ & - J(z-x) - J(z-y) - J(x+y-z). \end{aligned} \tag{1.5}$$

J is continuous on $[0,1]$, with $J(1-x) = -J(x)$. An ordinary boundary branch has profile $g + c$, where the constant c records the entire action already accumulated in that branch. Such constants will always be restored, rather than assigned independently to an edge.

Theorem 1.1 (The original finite packet)**.** *For every fixed strict expression there are a primitive rectangular wire window $\mathcal{W}$, a number $\eta > 0$ and a holomorphic finite packet $B_N(\boldsymbol{\tau}, s)$ with the following properties. The exact reduced source, with its actual fixed crossing powers, equals its circle trace functionals applied to B_N, with the original closing, reference and framing scalars and the single factor H_N, up to $N^A e^{N(F_0 - \eta)}$. Its exponential loss is independent of the magnitudes of the fixed crossing powers. Constants and starting levels may depend on those powers.*

The packet retains every original physical edge factor (1.3), every ordered complete bare gate and every $d = N - 1$ offset. Uniformly on a fixed complex angular neighborhood,

$$B_N(\boldsymbol{\tau}, s) = N^{3(1-H)/2}\Big(\prod_e \gamma_{e,N}(s)\Big) a(\boldsymbol{\tau}) e^{N\Psi(\boldsymbol{\tau})}(1 + O(N^{-1})), \tag{1.6}$$

where a is holomorphic and nonzero. The real action Ψ is separately even, $\Psi(0) = F_0$, and

$$\Psi_{\tau_v \tau_w}(0) = -\delta_{vw} a_v, \qquad a_v = \frac{1}{2\pi}\sum_{e \ni v} \tan(\pi x_e^*) > 0.$$

On a fixed strip about the whole real angular cube it obeys

$$|B_N(\mathbf{t} + i\mathbf{y}, s)| \le N^A e^{N(F_0 - c\|\mathbf{t}\|^2 + C\|\mathbf{y}\|^2)}, \qquad |t_v| \le \pi, \quad |y_v| \le y_0. \tag{1.7}$$

All statements hold on each fixed disk $|s| \le R < 1$ and on each fixed compact subset of the open upper half-plane, with fixed derivatives on smaller compacts and with both level parities.

The formulas for this packet and its lattice appear in Section 6. The source error is bounded before moving the circle contours. In particular its proof does not multiply relative errors at several independently filled local saddles.

Theorem 1.2 (The joint filled source)**.** *For each fixed strict knot expression there is a constant B such that all fixed signed powers $|n_v| > B$ have a nonzero joint first-charge asymptotic. No ratio between the $|n_v|$ is restricted. On the same meridian compacts as above,*

$$\frac{F_N(q^s)}{F_N(1)} = H_N(s)q^{a(D)s^2/2}(1+O(N^{-1})), \qquad a(D) = -w(D) - \sum_v n_v. \tag{1.8}$$

The leading coefficient before normalization is nonzero and its joint wire-and-angle Hessian is invertible. After the scalar in (1.8) is removed, the only meridian dependence in the retained local integral is the product over original physical edges of

$$\mathcal{I}_N(x_e, s) = \frac{G_N(Nx_e)^2}{G_N(Nx_e - s)G_N(Nx_e + s)}. \tag{1.9}$$

G_N is the canonical sine-factorial interpolation of [5, Lemma 2.1]; the exact affine edge index is used in (1.9). All other charges and endpoint currents have a fixed exponential loss relative to the joint first charge. The threshold B is chosen before the meridian compact.

The state F_N is the zero-framed reduced moving-meridian invariant of [1]; at $s = 0$ it is the reduced Kashaev value for $N \geq 3$. We prove an analytic source theorem. Geometric action and preferred-longitude identifications, and the extraction of all inverse-level coefficients, are separate consequences.

2 Uniform finite bounds with all normal data

For a canonical polynomial of degree less than N, let $\|F\|_{\mathrm{J}}$ be the largest absolute value at a distinct complete node and the largest first derivative at a double node. At odd levels -2 is simple and contributes only its value. Complete-node conditioning, [4, Lemma 2.1], proves polynomial equivalence of this norm, the interval norm and the Chebyshev coefficient sum. It also proves

$$\|e_r\|_{\mathrm{J}} \leq N^A w_r, \qquad w_r = \begin{cases} e_r(-2), & r \leq m, \\ 1, & r > m, \end{cases} \qquad \log w_r = Ng(r/N) + O(\log N) \quad (r \leq m). \tag{2.1}$$

The high polynomials have zero complete values but generally nonzero first normals. No argument below discards those derivatives.

Write $C_j(u+u^{-1}) = (u^j + u^{-j})/2$, with $C_0 = 1$. Their coefficient sum is submultiplicative, since $2C_iC_j = C_{i+j} + C_{|i-j|}$. In the quotient P_σ, $C_N = -b$, where $b = \cos(2\pi\sigma)$, and

$$C_{N+r} \equiv -2bC_r - C_{N-r}, \qquad C_{kN+r} \equiv -2bC_{(k-1)N+r} - C_{(k-2)N+r}.$$

For $r = 0$ start with $1, -b$. Thus reducing a polynomial of degree at most hN costs at most $(2|b|+1)^h$ in this norm. The integer h is bounded by the fixed vertex valence. Since the top monomial coefficient of C_d is $1/2$, both the full trace and fixed-valence quotient products have polynomial cost in $\|\cdot\|_{\mathrm{J}}$. This proves these statements at collisions as polynomial identities.

Lemma 2.1 (Physical coupons on a fixed microscopic chart)**.** *Every fixed signed crossing power has polynomial canonical norm, uniformly on $|s| \leq R < 1$ and on fixed nonreal compacts. The same holds after multiplication by any ordinary circle polynomial $L_j = (-1)^{j-1}U_{j-1}(z/2)$, $1 \leq j \leq N$. Every typed turned coupon with input E has coefficients bounded by*

$$|[e_k]\mathcal{T}(pE)| \leq CN^A \|E\|_{\mathrm{J}}\, w_k. \tag{2.2}$$

Constants may depend on the fixed word and compact, not on N, j, k.

Proof. For an equal-color power the full crossing polynomial is $\kappa_\sigma^n r_{N,\sigma}^{[n]}$, where

$$\kappa_\sigma = q^{(d+\sigma)^2/2}, \quad t_j = q^{-2j-1+2\sigma}, \quad c_j(\sigma) = t_j + t_j^{-1}, \quad \lambda_j = (-1)^j q^{j(j+1)-2\sigma j}.$$

Proposition 6.1 of [1] identifies its values and full collision normals. Away from $U - U^{-1} = 0$, $U = e^{2\pi \mathrm{i}\sigma}$, its exact polynomial interpolation is

$$r_{N,\sigma}^{[n]}(z) = -\frac{1}{N(U-U^{-1})} \sum_{j=0}^{N-1} \lambda_j^n (t_j - t_j^{-1}) \frac{P_\sigma(z)}{z - c_j(\sigma)}. \tag{2.3}$$

The last quotient is a divided difference of T_N. The nodes lie in a Bernstein ellipse of angular height $O(N^{-1})$; the first two derivatives of T_N there are $O(N^2)$ and $O(N^4)$. Integrating along a segment from a real point to a node bounds both the polynomial and its derivative by CN^4 on a compact avoiding the displayed denominator. All $\lambda_j^{\pm n}$ and $\kappa_\sigma^{\pm n}$ have bounded modulus there.

Inside $|\sigma| < 1$, colliding distinct nodes occur at $0, \pm 1/2$. Their indices satisfy $j+l+1-2\sigma = N$, whence

$$\lambda_l/\lambda_j = (-1)^{l-j} q^{(l-j)(j+l+1-2\sigma)} = 1.$$

The node difference has a simple zero and the value difference vanishes. The divided differences and the confluent interpolation extend holomorphically. A larger parameter circle of radius less than one, different from $1/2$, and the maximum principle give the same polynomial bound throughout each smaller disk. This also handles negative powers. Nonreal compacts use (2.3) directly. Mixed words are the fixed powers of the constant-channel crossing, times the exact monodromy phases in [1, Section 8]; their norms have the same bound. Since $|U_{j-1}(z/2)| \le j$ on $[-2, 2]$, polynomial conditioning and quotient reduction prove the assertion for p.

For completeness the paired factors have uniformly bounded upper and inverse bounds on these charts. Euler's product gives

$$W_k(s) = \prod_{\substack{1 \le r \le k \\ a \in \mathbb{Z}}} \left(1 - \frac{s^2}{(r+aN)^2}\right). \tag{2.4}$$

The integers appearing here are distinct and nonzero. Outside a fixed finite set the logarithms are bounded by $C/|r+aN|^2$. The finite remaining factors stay away from zero on the stated compacts. Products of their minima with one and maxima with one bound every subset uniformly. Also H_N converges uniformly to $\sin(\pi s)/(\pi s)$, nonzero there; the value at zero is removable. The exact identities $W_k W_{d-k} = H_N^2$ and

$$[e_k]\mathcal{T}(pE) = \gamma \frac{W_{d-k}}{H_N} \phi_\sigma(e_k p E) \tag{2.5}$$

now prove (2.2) by the full norm and quotient estimates. □

A coefficient loss in E is inherited linearly in (2.2); there is no substitution of independent branch maxima. Conversely a bare turned output will be estimated by its actual complete probe profile, rather than repeatedly charging the coarser cost $2 \max g$.

3 Whole traces and reflected high indices

The following comparisons concern the actual top-coefficient trace. They retain all its Fourier aliases, including the alternating signs at even levels. Put $\epsilon = (-1)^{N+1}$.

Lemma 3.1 (The finite Fourier trace)**.** *If* $F(-u-u^{-1})=\sum a_k u^k$*, then*

$$\phi_s(F)=N^2\sum_{l\geq 1}\epsilon^{l+1}(a_{lN-1}-a_{lN+1})U_{l-1}(\cos 2\pi s). \tag{3.1}$$

Only finitely many summands occur. The formula holds at every collision with the full normal functional.

Proof. At generic s choose $u_j^N=\epsilon e^{2\pi \mathrm{i}s}$ and put $z_j=-u_j-u_j^{-1}$. These are the distinct roots of P_s, and

$$P_s'(z_j)=\frac{2\mathrm{i}N\sin 2\pi s}{u_j-u_j^{-1}},\qquad \phi_s(F)=\frac{\epsilon N}{2\mathrm{i}\sin 2\pi s}\sum_j (u_j-u_j^{-1})F(-u_j-u_j^{-1}).$$

The root sum selects multiples of N. Pairing opposite exponents gives (3.1), since $a_{-k}=a_k$. Both sides are polynomials in $2\cos 2\pi s$: the left by monic reduction, the right by the finite Chebyshev sum. Polynomial continuation proves the collision assertion, including all first normals. □

For low n, the exact coefficients are positive:

$$[u^k]e_n(-u-u^{-1})=\frac{S_{2n}}{S_n^2 S_{n+k}S_{n-k}},\qquad |k|\leq n. \tag{3.2}$$

Indeed the numerator is $u^{-n}\prod_{a<n}(1+q^{2a+1}u)(1+q^{-2a-1}u)$. Expand the consecutive geometric factors by the finite binomial identity, proved by multiplying by its last factor. The phases cancel and give (3.2). Every factorial index is less than N. Symmetry is explicit; the successive ratio on the nonnegative side is $\sin((n-k)\pi/N)/\sin((n+k+1)\pi/N)\leq 1$. Thus these sequences are symmetric and unimodal. Their convolutions are too: write a symmetric unimodal sequence as a nonnegative sum of centered interval indicators.

We will also use an elementary quantitative tail fact. If a is the convolution of B nonnegative symmetric unimodal sequences supported in $[-m,m]$, then

$$a_k\leq (2m+1)^B(a_k-a_{k+2}),\qquad k\geq m. \tag{3.3}$$

To prove it after the interval decomposition, let C be the convolution of integer interval indicators, with largest radius r. Write $C=\mathbf{1}_{[-r,r]}*H$. For $k\geq r$, $C(k)-C(k+1)=H(k-r)-H(k+r+1)$. If $C(k)>0$, the first argument is in the support. If the second is outside, the difference is positive. Otherwise induction makes its last step strict, since $k+r$ is at least every radius in H; unimodality handles the other steps. The one-factor case is immediate. The positive integer difference is at least one, while $C(k)\leq (2m+1)^B$. Sum the interval decompositions to obtain (3.3).

Proposition 3.2 (Same-level low trace comparison)**.** *For fixed* B *and any bounded set of complex* s*,*

$$|\phi_s(e_{n_1}\cdots e_{n_B})|\leq CN^A\phi_0(e_{n_1}\cdots e_{n_B}),\qquad n_i\leq m. \tag{3.4}$$

A zero on the right forces the left to vanish. Both parities are included, and all factors on the right are at the original level.

Proof. Write $D=\sum n_i$ and a_k for the product coefficients. All $d_l=a_{lN-1}-a_{lN+1}$ are nonnegative. The number of aliases in (3.1) is at most B, and

$$|U_{l-1}(\cos 2\pi s)|\leq l e^{2\pi(l-1)|\Im s|}.$$

For odd N this proves the result directly, since all complete alias signs are positive. If $D < N-1$, both traces vanish by degree. It remains to consider even $N \geq 4$, $D \geq N-1$.

Put $L = N-1$, an odd level; its low bound is the same $m = (N-2)/2$. Concavity of sine and reflection about $\pi/2$ give, for $0 \leq r \leq N-2$,

$$\left(\frac{N}{N-1}\right)^r \frac{N-1-r}{N-1} \leq \frac{S_r^{(L)}}{S_r^{(N)}} \leq \left(\frac{N}{N-1}\right)^r. \tag{3.5}$$

For the lower estimate multiply $\sin(\pi a/(N-1))/\sin(\pi a/N) \geq N(N-1-a)/((N-1)(N-a))$; the product telescopes. The upper estimate follows from decreasing $\sin t/t$. All these ratios and their inverses are polynomial in N. Changing an allowed factorial index by one also costs at most a polynomial, since $4/N \leq 2\sin(\pi a/N) \leq 2$. Consequently (3.2) and convolution give $a_k^{(N)} \leq N^A a_k^{(L)}$ at every index, with exactly the same support. Unimodality and (3.3), now at $k = lL-1 \geq m$, imply

$$d_l^{(N)} \leq N^A a_{lN-1}^{(L)} \leq N^A a_{lL-1}^{(L)} \leq N^{A+B}(a_{lL-1}^{(L)} - a_{lL+1}^{(L)}).$$

The odd complete trace therefore bounds the absolute even trace up to a polynomial.

We must return to the original level. The complete multiplication law, [2, Theorem 3.3], is positive on

$$\mu_K(i,j;r) = \frac{S_{i+j}^{(K)} S_r^{(K)}}{S_i^{(K)} S_j^{(K)} S_{r-i}^{(K)} S_{r-j}^{(K)} S_{i+j-r}^{(K)}}, \quad i+j<K, \quad \max(i,j) \leq r \leq i+j; \tag{3.6}$$

it is zero otherwise. A successive odd-level trace path starts at $r_1 = n_1$, ends at $r_B = N-2$, and has nonnegative gate slacks $r_{j-1} + n_j - r_j$. Their sum is $D-(N-2) \geq 1$. At its first positive slack raise that output and every subsequent running output by one. At the first gate the slack permits the increase; at later gates the input sum and output both increase by one, so all support inequalities persist and input sums remain at most $N-1$. The new path closes at $N-1$, the even trace target. Every image has at most $B-1$ preimages, since specifying the first changed gate recovers the path. Each factorial index changes by at most one. Equation (3.5) thus bounds each old positive path weight by a polynomial times its image weight. Summing proves $\phi_{L,0} \leq N^A \phi_{N,0}$, completing the proof. This level comparison is only an estimate, not an identification of invariants at different levels. □

Proposition 3.3 (A reflected high trace)**.** *Let $h > m$, $r = d-h$ and $F = \prod_{j=1}^B e_{n_j}$, $n_j \leq m$. For fixed B and bounded s,*

$$|(\phi_s - \phi_0)(e_h F)| \leq C N^A |\Delta_s|\, \phi_0(e_r F), \qquad \Delta_s = 2\cos 2\pi s - 2. \tag{3.7}$$

If $n_j/N \in [\delta, 1/2-\delta]$ and $r/N \geq \delta > 0$, the right side can be multiplied by $e^{-\kappa N}$, with $\kappa > 0$ independent of the meridian compact.

Proof. Set $a = N-h$, $v = 2h-N \geq 0$. Removing one complete cycle of geometric factors from the Newton numerator gives

$$\prod_{j<h}(1+q^{2j+1}u)(1+q^{-2j-1}u) = (1-\epsilon u^N)\prod_{j=0}^{v-1}(1-q^{2j-v+1}u).$$

The last product has coefficient $(-1)^l S_v/(S_l S_{v-l})$. Thus the Laurent coefficients A_k of $e_h(-u-u^{-1})$ are zero for $|k| < a$, and otherwise

$$A_k = (-1)^{h-|k|} \frac{S_{2h-N}}{S_h^2 S_{|k|-a} S_{h-|k|}}, \qquad a \leq |k| \leq h. \tag{3.8}$$

At even $h = N/2$ the only coefficients are $A_h = A_{-h} = S_h^{-2}$. By (2.1) and polynomial conditioning, the interval norm of this high polynomial is polynomial; its Laurent coefficients are cosine coefficients, so $\sum |A_k| \le N^A$.

Write $F(-u - u^{-1}) = \sum f_k u^k$. Its coefficients are positive and unimodal. Alias one in (3.1) is independent of s and cancels in the difference. All others have frequencies at least $2N - 1$, so convolution gives a bound by $N^A f_{2N-1-h} = N^A f_{N+r}$. Moreover $U_{l-1}(\cos 2\pi s) - l$ is divisible by Δ_s with bounded quotient for the fixed finite set of aliases. This proves

$$|(\phi_s - \phi_0)(e_h F)| \le C N^A |\Delta_s| f_{N+r}. \tag{3.9}$$

If $C = e_r F = \sum c_k u^k$, the endpoint coefficient of e_r is S_r^{-2}, and $S_r^2 \le N$ on the low range. Hence

$$c_{N-1} \ge S_r^{-2} f_{N-1-r} \ge N^{-1} f_{N+r}.$$

The tail lemma bounds c_{N-1} by a polynomial times $d_1 = c_{N-1} - c_{N+1}$. To bound this first difference by the complete trace even when its aliases alternate, form $Q(X) = \sum_l \epsilon^{l+1} d_l U_{l-1}(X)$. Its degree is bounded by B. Evaluate at $B+1$ fixed distinct values $X_j = \cos 2\pi s_j$, $0 \le s_j < 1/8$. The inverse evaluation matrix is fixed, so $d_1 \le C \max_j |Q(X_j)| \le N^A \phi_0(C)$ by Proposition 3.2. This proves (3.7) at the same original level.

For strictness retain the two different frequencies above. Define

$$b_u(t) = J(2u) - 2J(u) - J(u + t) - J(u - t), \quad |t| \le u.$$

On $u \in [\delta, 1/2 - \delta]$,

$$b_u''(t) = -\frac{\pi \sin(2\pi u)}{\sin \pi(u + t) \sin \pi(u - t)} \le -\lambda_\delta < 0.$$

The constrained convolution action $\mathcal{B}(t) = \max_{\sum t_j = t} \sum b_{u_j}(t_j)$ is even and strongly concave with constant λ_δ / B, by Cauchy–Schwarz on the coordinate differences of two maximizers. It follows that

$$\mathcal{B}(t_2) - \mathcal{B}(t_1) \le -\frac{\lambda_\delta}{2B}(t_2^2 - t_1^2), \quad 0 \le t_1 \le t_2. \tag{3.10}$$

Uniform sine-factorial estimates give $\log f_k = N\mathcal{B}(k/N) + O(\log N)$ on the support. For the lower bound round a maximizing tuple, preserving its integer sum: floor all coordinates and raise the required number with nonzero fractional parts. The integer endpoints remain admissible. The modulus $|J(x + h) - J(x)| \le C|h| \log(e/|h|)$ controls even endpoint rounding; the upper bound counts polynomially many tuples. Apply (3.10) at $(N - 1 - r)/N$ and $(N + r)/N$. When the latter coefficient is nonzero,

$$f_{N+r} \le N^A e^{-2\lambda_\delta r/B + O(1)} f_{N-1-r}.$$

Since $r \ge \delta N$, this supplies the asserted strict factor in the preceding comparison. If it is zero, the difference is zero. □

Corollary 3.4 (Regular low windows)**.** *If all $n_j/N \in [\delta, 1/2 - \delta]$ and $\sum n_j/N > 1 + \delta$, then $\phi_s(F)/\phi_0(F) = 1 + O(e^{-\kappa N})$ uniformly on any fixed meridian compact, with fixed derivatives. For at most four low factors the traces agree exactly, without a window assumption.*

Proof. Use the same convolution action at $t_1 = 1 - 1/N$ and $t_2 = l - 1/N$, $l \ge 2$. Their squared difference is at least two. The tail lemma gives $d_1 \ge N^{-B} a_{N-1} > 0$, so the sum of all other d_l is exponentially smaller. Formula (3.1) at s and at zero proves the assertion, including the even signs. With four low factors the raw degree is at most $4m \le 2N - 2$, so there is no second alias. Cauchy's estimate on a larger fixed meridian compact proves the derivative statement. □

4 Positive pressures with shared boundaries

The estimates just proved have different complete tensors on their right sides. This section compares their entire contractions. The comparison does not assert an isotopy of the corresponding tangles.

Let $\mathcal{G}$ be a finite tree of bare junctions, each of full valence at least three. Its internal edges are physical turns and its ordinary external profiles are $g + c_j$. A covariant junction label is the index inserted in its trace; its rooted ordinary output is the complementary label. Thus a physical edge identifies equal covariant labels, while a hidden unturned resolution edge identifies complementary ones. The cubic action is symmetric:

$$t(x,y,z) = m(x,y,1-z) = J(x+y)+J(x+z)+J(y+z)-J(x)-J(y)-J(z)-J(x+y+z-1). \tag{4.1}$$

Choose an ordered binary resolution at each junction. Let $V_D(a_1,\dots,a_D)$ denote the maximum of its gate action with the covariant boundary labels fixed, and let $F(\mathcal{G})$ be the maximum after including all ordinary profiles and shared edges.

Lemma 4.1 (Strict resolutions and constrained maxima)**.** *The strict support of a whole junction is*

$$a_i > 0, \qquad a_i + a_j < 1 \ (i \neq j), \qquad \sum_i a_i > 1. \tag{4.2}$$

Every prescribed binary resolution realizes that support. The tree pressure has a unique strict maximum. The same assertion holds if at most one external incidence of each whole junction is fixed at $1/2$*, including small independent variations of those fixed labels. The constrained pressure is analytic there with negative definite Hessian in its free boundary labels.*

Proof. For a product of inputs a_i, its strict output interval is

$$\max_i a_i < b < \min(1, \sum_i a_i), \qquad a_i + a_j < 1.$$

One proves this for any binary association by induction. A composite child has output interval (L_S, U_S) with $L_S = \max_{i\in S} a_i$, $U_S = \min(1, \sum_{i\in S} a_i)$; a singleton is fixed. Choose the two child outputs below b with sum between b and one. Their lower sum is below one by the pair condition and their upper sum exceeds b. The singleton cases use the same inequalities directly. Conversely the gate at which two inputs first meet forces their pair sum below one. Closing the last incidence makes $b = 1 - a_D$ and gives (4.2). Its closed version follows by the same finite interval rules.

Assign $3/8$ to every physical edge and ordinary boundary. This is a strict witness at every junction. If one boundary at a junction is fixed at half, its largest pair is $7/8$ and its total is at least $5/4$; otherwise these bounds are $3/4$ and $9/8$. The witness persists under small independent variations of the fixed halves.

Here the endpoint argument must also retain those fixed boundaries. For positive triples of sum one put $\mathcal{V}(a,b,c) = -\pi(J(a) + J(b) + J(c))$. Its Hessian is negative definite on the angle plane: the diagonal cotangent sums are positive and $\cot\pi a \cot\pi b + \cot\pi b \cot\pi c + \cot\pi c \cot\pi a = 1$. If $p = z - x$, $q = z - y$, $r = x + y - z$, $w = 1 - x - y$, then

$$\pi m(x,y,z) = \mathcal{V}(y,x,w) + \mathcal{V}(p,q,2r+w) + \mathcal{V}(p+q,r,r+w). \tag{4.3}$$

The first atom's zero quadratic form forces $dx = dy = 0$ and the second then forces $dz = 0$. Thus a strict gate has a negative definite Hessian in all three variables. Also $g'(a) = 2\log\cot\pi a$ and $g'' < 0$ on $(0, 1/2)$.

At a nonvertex flat triple $(0, a, 1-a)$ an inward segment has derivative $+\infty$, because $J(tu) = tu \log t + O(t)$. At a vertex flat triple the $t \log t$ terms cancel, so no negative infinite term can cancel this barrier. Root the resolved tree at an output or a fixed half incidence. Output zero forces both product inputs to zero. Output one either has two interior inputs summing to one, which supplies a nonvertex flat atom, or propagates an endpoint to a child. A physical turn exchanges zero and one; an unturned resolution edge preserves the ordinary endpoint. Following this descent in a finite tree reaches an ordinary leaf endpoint, where the inward derivative of g is infinite, or an interior pair summing to one. It cannot terminate at an interior fixed boundary. No maximizer therefore has an endpoint wire. With all wires interior, any remaining equality in the gate support supplies a nonvertex flat atom in (4.3). The maximum is strict.

Every free coordinate appears in a strict gate or a leaf term, so the summed Hessian is negative definite. Concavity gives uniqueness. Including the variable prescribed boundaries in that quadratic form and taking a Schur complement proves the asserted strict concavity of their pressure. The implicit function theorem and compactness give the analytic local maximizing branch. □

Lemma 4.2 (Reflection and deletion of turns)**.** *An unturned rooted side with ordinary boundaries has profile f on $(0,1)$ satisfying $f(b) \geq f(1-b)$ for $b \geq 1/2$. The inequality is strict above half unless the entire rooted side is one trivalent junction with two ordinary leaves. Consequently deleting any set S of physical turns and fusing their adjacent ordered junctions gives*

$$F(\mathcal{G}/S) \leq F(\mathcal{G}). \tag{4.4}$$

Every internal physical label at the original maximum is at most half. Equality occurs only for the one internal edge of the entire two-trivalent-junction tree with two ordinary leaves on each side.

Proof. We use the stationary response equations of [3, Lemma 4.1 and Proposition 4.2], recording their inequality consequence to preserve the constants. Put

$$B = \cot \pi b, \quad \kappa = e^{-f'(b)}, \quad X^2 = \frac{1+B^2}{1+\kappa}, \quad Y^2 = \kappa X^2.$$

An ordinary leaf has $(B, X, Y) = (B, B, 1)$. A composite has $-Y < B < X$. At a product the inputs and output have common X and

$$B_o = \frac{B_1 B_2 + X^2 - 1}{B_1 + B_2}, \qquad Y_o = \frac{Y_1 Y_2}{B_1 + B_2}, \qquad B_1 + B_2 > 0. \tag{4.5}$$

A physical turn sends (B, X, Y) to $(-B, Y, X)$. Proper responses remain proper. To check the possible signs directly, for $X \geq 1$ the input inequalities force $B_i > 0$ and $B_o - X = ((X - B_1)(X - B_2) - 1)/(B_1 + B_2) < 0$. For $X < 1$, write $B_i = \sqrt{1-X^2} \cot \alpha_i$, $Y_i = \sqrt{1-X^2} \csc \alpha_i$; the denominator gives $\alpha_1 + \alpha_2 < \pi$ and the product response has this angle. The asserted inequalities follow. Thus these formulas also apply to every turned composite side in the present tree.

At the last gate of an unturned product,

$$Y^2 = (B_1 - B)(B_2 - B) \leq (X - B)^2, \qquad X \geq X_2(B) := \frac{B + \sqrt{B^2 + 2}}{2}.$$

The factors are positive by strict support. Since $\log(X^2/(1 + B^2 - X^2))$ increases with X,

$$f'(b) \geq f_2'(b) = 2 \log(2 X_2(B)^2), \tag{4.6}$$

strictly when either input is composite. The actual two-leaf profile f_2 is symmetric, because $X_2(B)X_2(-B) = 1/2$. Integrating (4.6) from $1-b$ to b proves the reflection statement, with all additive constants unchanged. The exceptional side is precisely the two ordinary leaves. At $b = 1/2$ its derivative is zero; every other side has positive derivative there.

Across an original edge the two ordinary outputs are both $1-z$, so its cut pressure is $f_A(1-z) + f_B(1-z)$. Above half it decreases strictly; at half it decreases unless both sides are exceptional. This proves the internal-label statement. The boundary labels themselves remain strictly between zero and half.

Exact fusion is supplied by the full metric:

$$\sum_{i=0}^{d} \phi_0(Ae_i)\phi_0(e_{d-i}B) = \phi_0(AB). \tag{4.7}$$

Indeed expand $A = \sum_i \phi_0(Ae_i)e_{d-i}$. Reflecting one endpoint of an equal-index physical edge gives exactly this complementary pairing. The induced order is the boundary order of the two vertex disks and their joining band; it substitutes the child product in its original position. No attached branches are permuted. The pressures before and after this fusion are

$$F_T = \max_b (f_A(b) + f_B(b)), \qquad F_I = \max_b (f_A(b) + f_B(1-b)).$$

For $b \geq 1/2$ use reflection on B, and for $b \leq 1/2$ use it on A at $1-b$. Thus $F_I \leq F_T$. A fused junction has valence $D_1 + D_2 - 2 \geq 4$, retains ordinary boundary constants, and remains in the same class. Apply this argument successively to the actual profiles of each current fused tree to obtain (4.4). Equality is possible; no strict source loss has yet been inferred. □

Lemma 4.3 (Several high boundaries)**.** *Fix marked covariant boundary labels $z_i \geq 1/2$, at most one at each retained whole junction, without leaf actions on those marks. Let $\Psi(\mathbf{z})$ be the maximum of the same complete action. Then $\Psi(\mathbf{z}) \leq \Psi(\frac{1}{2}, \dots, \frac{1}{2})$. Moreover, if a whole junction has $k \geq 2$ half incidences, then*

$$V_D(\underbrace{\tfrac{1}{2}, \dots, \tfrac{1}{2}}_{k}, x_1, \dots, x_{D-k}) = \sum_j g(x_j), \qquad 0 \leq x_j \leq \tfrac{1}{2}. \tag{4.8}$$

Proof. Lemma 4.1 gives a strict analytic constrained maximum at the all-half point. Root there at mark i. Each other fixed half terminal has $B = 0$ and the positive response $\kappa = e^{m_x}$ supplied by its incident strict gate; hence it is proper, $-Y < B < X$. This response is not a new leaf action. Propagate it by (4.5). The final product has ordinary output half and two inputs below half, so $B_1, B_2 > 0$ and

$$Y^2 = 1 - X^2 = B_1 B_2 \leq X^2, \qquad \partial_{z_i} \Psi = -\log(X^2/Y^2) \leq 0.$$

The symmetric cubic action (4.1) shows that rerooting changes no constrained pressure. Its joint concavity therefore gives the claimed simultaneous bound by the supporting plane at the all-half point. Closed boundary states follow by continuity. No assertion of coordinate monotonicity with other marks already above half is needed.

For the second assertion use odd levels $N = 2m+1$ solely to calibrate the real action. The polynomial e_m^2 is the full Hermite projector at the simple node -2: it vanishes with both jets at every double node, and

$$e_m(-2) = 1, \qquad \phi_0(e_m^2) = 1, \qquad \phi_0(e_m^2 F) = F(-2).$$

For example $S_m^2 = N$ and $P_0'(-2) = N^2$ give the first value; the top coefficient of e_m^2 gives the second. Hence $\phi_0(e_m^k \prod e_{n_j}) = \prod e_{n_j}(-2)$ for $k \geq 2$. Take its uniform closed factorial

rate, using the positive ordered resolution (6.1). Here is a support-preserving approximation, including the saturated pair faces. Keep every half index exactly $m = d/2$ and set the other inputs to $n_j = \lfloor dx_j \rfloor$. All pair sums are at most d, and the two retained halves make the total at least d. The integer interval rules of the complete product therefore give at least one supported resolution. In a sequential resolution its running outputs obey only integer difference inequalities: $r_a - r_{a-1} \in [0, n_a]$, $r_a \geq n_a$, $r_{a-1} \leq d - n_a$, with the fixed first and last outputs. Encode bounds by a further coordinate $r_0 = 0$.

For an arbitrary real supported resolution let Y_a be its outputs multiplied by d, and $Y_0 = 0$. Every rounded difference inequality $r_j - r_i \leq b_{ij}$ is violated by Y by at most two. The integer system is feasible, so its directed constraint graph has no negative cycle. Let d_{ij} be its shortest-path lengths; all coordinates are bounded, so these lengths are finite. A shortest path can be taken simple, with at most D edges, giving $d_{ij} \geq Y_j - Y_i - 2D$. The integer vector $z_j = \min_i(\lfloor Y_i \rfloor + d_{ij})$ satisfies every difference inequality and $Y_j - 2D - 1 \leq z_j \leq Y_j$. Subtracting z_0 fixes the zero coordinate without changing any inequality. The resulting supported integer outputs differ from Y by at most $2D + 1$ and satisfy both prescribed endpoint outputs exactly. Thus every real resolved state has supported lattice approximants within $O(1/d)$. Conversely every lattice state is a point of the same limiting closed support. Uniform closed factorial estimates give $O(\log N)$ errors for these bounded displacements. Positive state counting then identifies the exact finite rate with the maximum action. This proves (4.8) in its relative interior, and continuity adds zero arms and further half arms. The real action is independent of parity; no even-level simple-node projector is asserted. □

These two lemmas give a useful reassembly principle. Remove a nonempty collection of pairwise nonadjacent moving junctions, each with at least two high incidences, and bound each by the sum of g on its low incidences and zero on the high ones. The neighboring retained complete hubs have at most one high incidence by their pairwise support. Mark those high boundaries and charge a single g to each low arm of a removed hub. Optimize each coupled old region, move all its marks to half by Lemma 4.3, then reinsert the removed hubs by (4.8). Their cardinal action consumes the charged g profiles exactly once. Original low external profiles $g + c_j$ remain in addition; original high external arrays keep their constant c_j and acquire $g(1/2) = 0$. This gives a feasible state of the original complete tree with at least one pair sum equal to one. That finite union of closed boundary faces misses the unique strict maximum. Compactness therefore supplies a fixed strict gap. This argument also treats a single removed hub with only external arms and all closed endpoints.

5 The whole bare component

An estimate of a tensor with one high incidence cannot generally use the complete tensor at that same index. The reflection in Lemma 3.3 changes the pairing across that edge. We now keep that change throughout the neighboring component.

Theorem 5.1 (Component comparison)**.** *Let A be a fixed unturned bare component, and $Q = \mathcal{T}(A)$ its typed output. Suppose its incoming full coefficient arrays obey*

$$|f_{j,r}| \leq e^{N(c_j + o(1))} \begin{cases} e^{Ng(r/d)}, & r \leq m, \\ 1, & r > m, \end{cases} \tag{5.1}$$

uniformly on a stated meridian compact. Let A^0 be its original complete positive reference with ordinary boundary profiles g, and put $\Omega = \Phi_{C(hA^0)} + \sum_j c_j$, using the actual output probe of [3, Theorem 4.4]. Then

$$Q = Q^{\mathrm{clean}} + E, \qquad \|Q\|_{\mathrm{J}} \leq e^{N(\Omega + o(1))}, \qquad \|E\|_{\mathrm{J}} \leq e^{N(\Omega - \eta + o(1))}$$

for some $\eta > 0$. In the clean term all bare tensors are complete, all original external inputs and the output are low, and every complete-supported internal index is retained. Actual low arrays, physical edge weights and typed scalars are retained. A loss in one entire incoming array is inherited without change.

Proof. Expand each moving vertex as $V_\sigma = V_0 + \delta V_\sigma$, then partition every index into its low and high bands. There are only finitely many patterns. Unselected and constant vertices retain their positive complete tensors. The complete pair support permits at most one high incidence at each such vertex.

For a selected difference use the all-low estimate of Lemma 3.2, the one-high reflected estimate of Lemma 3.3, or the full norm bound

$$|\delta\phi_\sigma(\prod_i e_{n_i})| \le N^A \prod_{n_i \le m} w_{n_i}. \tag{5.2}$$

The last inequality follows from (2.1), fixed-valence quotient reduction and the polynomial trace norm. It retains the high normals, even when all their values vanish.

Consider a selected one-high difference whose high incidence is internal. Its neighboring vertex is constant because moving equal-color vertices are nonadjacent. Each endpoint has only this one high incidence. Hence all such selected edges form a matching S. Reflect h to $r = d - h$ at the moving endpoint and use (4.7). Summing the complementary pair fuses the two whole vertices in their induced order. Removing its band restriction only enlarges the positive sum. Every fused vertex has valence at least four and low exterior incidences. The restored complementary label remains one hidden unturned wire in a positive resolution. No other branch is replaced by an independent maximum. By Lemma 4.2 the resulting pressure is at most the original pressure. For a one-high external incidence there is no fusion: reflection makes its array rate c_j while the low reference has rate $c_j + g(r/d)$. Keep the nonnegative surplus $g(r/d)$.

Selected vertices with several high incidences are not fused. They remain nonadjacent. Remove them using (5.2), assigning one g to each low arm and marking every high arm in the retained old region. Mark also an unreflected high original input or output. Every retained whole vertex has at most one mark; the matching fusions did not create high exterior incidences. The reassembly following Lemma 4.3 moves these marks simultaneously to half and reinserts the removed vertices. Each charged g is consumed once by their cardinal action. All original constants remain once. The result is a boundary state of the same fused complete tree: either a pair sum is one or an ordinary external leaf is half. Its unique strict maximum avoids this closed set, so it has a fixed loss. This includes a component consisting of a single removed vertex. With no removed vertex the same half-boundary argument gives the high-input or high-output loss directly.

It remains to consider a nonempty difference pattern with no such boundary. Its positive comparison is a fused tree $\mathcal{G}/S$. Outside a small neighborhood of its unique strict resolved maximum, compactness gives a fixed loss. Inside that neighborhood there are three possibilities. An externally reflected label lies strictly between zero and half, so its saved g is strictly positive. If an internal reflection occurs, the fused tree has a vertex of valence at least four. It cannot be the exceptional entire two-trivalent-vertex tree of Lemma 4.2. Every physical exterior label of that fused pair is therefore in a compact low interval. Its restored wire r/d is strictly positive. A maximum above half is separated from the sector $r \le m$; otherwise the strict part of Lemma 3.3 applies, including at a limiting half wire. Finally, with no reflection, every selected difference is all-low in the original tree. The strict whole-vertex support gives a positive total slack. Lemma 3.4 then gives a loss. In the two-trivalent-vertex exception the raw cubic degree on the entire strict neighborhood is less than $2N - 1$, so its trace difference is exactly zero instead.

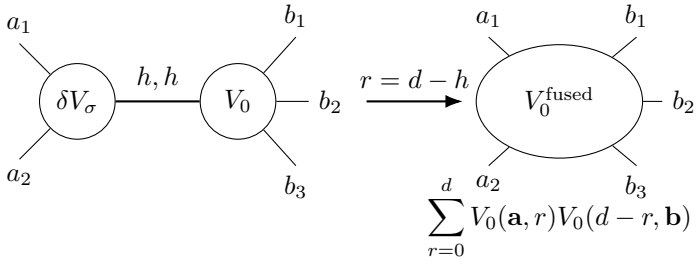


Figure 1: A selected high incidence joins a moving trace (left disk) to a constant trace. The original covariant labels agree, h, h. Reflection changes only the left label to $r = d - h$. The complete metric then pairs r with $d - r$ and fuses the disks in their boundary order. Every other arm remains in the same contraction. The picture depicts the tensor calculation used for an upper bound.

These cases exhaust the nonempty difference patterns. One strict factor is enough; all other factors retain their polynomial majorants. Uniform factorial estimates on the closed support polytope give $O(\log N)$ errors and only polynomially many states. Take the minimum of the finitely many positive losses and decrease it once to absorb those errors.

For the output use the exact identity (2.5):

$$\|Q\|_{\mathrm{J}} \le C N^{A} \sum_{r=0}^{d} w_r |\phi_\sigma(e_r A)|.$$

This is precisely the ordinary low probe g and the zero-rate high probe already treated. Its action is Ω, by [3, Theorems 4.3–4.4]; it is not a newly inserted physical circle. In the empty difference pattern with low external indices, the positive complete coefficients and bounded edge weights give the original positive reference, including its full low-output profile. Every internal complete-supported index stays in that term. This proves the asserted decomposition. Multilinearity preserves a loss in one full array in all comparisons, and shifts the corresponding constant c_j without changing any optimizer. □

We explain how this component statement is used in the original tree. A circle vertex, including its fixed signed crossing and typed turn, converts the actual incoming spectral norm into (5.1) by Lemma 2.1. A maximal bare component uses Theorem 5.1 and preserves its full clean low profile. Thus a small spectral-norm error is converted to a coefficient cone at a circle interface, where the exact extractor proves that conversion. It is never converted by an unsupported inversion of the Newton coefficient matrix inside a bare component. The mutual induction terminates because the original reduced graph is a finite tree. The root budgets of [3, Corollary 4.5] identify its accumulated pressure with F_0, retaining every additive constant.

This also gives fresh angular localization at any exposed circle. The clean output-probe profile is $f_A(1 - u) + g(u) + \sum c_j$, with a unique maximizing $u \in (0, 1/2)$. Outside a compact interval about it there is a strict loss. Within it the low Newton angular estimate [4, Lemma 7.2] gives a negative quadratic angular term. Errors have a strict spectral-norm gap, so complete Hermite conditioning and the degree-d Bernstein bound absorb them on a sufficiently small complex strip. This uses the actual accumulated pressure and works for an entirely bare-rooted expression by exposing any original circle through the typed adjunction. For $H = 2$ the same exact pairing is the two-leaf pure source; no nonexistent bare component is inserted.

6 The original finite wire packet

We now construct the packet in Theorem 1.1. Let J_b be the number of bare vertices and $E = H + J_b - 1$ the number of physical edges. Denote the sums of the circle and bare valences by D_c, D_b, so $D_c + D_b = 2E$.

The exact positive product coefficient, from the finite Newton calculus [2, Theorem 3.3], is

$$\mu_{ij}^{r} = \frac{S_{i+j} S_r}{S_i S_j S_{r-i} S_{r-j} S_{i+j-r}}, \qquad i + j \le d, \quad \max(i, j) \le r \le i + j, \tag{6.1}$$

and is zero outside this support. At a bare vertex with ordered labels $n_1, \dots, n_D$, put $r_1 = n_1$, $r_{D-1} = d - n_D$. Associativity within this one polynomial algebra gives

$$\phi_0\Big(\prod_{a=1}^{D} e_{n_a}\Big) = \sum_{r_2, \dots, r_{D-2}} \prod_{a=2}^{D-1} \mu_{r_{a-1}, n_a}^{r_a}. \tag{6.2}$$

The last pairing with e_{n_D} has coefficient one. There are $D - 3$ hidden variables and $D - 2$ gates, not an extra closing variable. The physical edge labels and these hidden labels are independent integer coordinates before inequalities are imposed. Their number and the gate count are

$$K = E + D_b - 3J_b, \qquad G = D_b - 2J_b, \qquad -D_c - \tfrac{3}{2}G + \tfrac{1}{2}K = \tfrac{3}{2}(1 - H). \tag{6.3}$$

All other indices are integral affine functions of them, with the literal complements $d - n = N(1 - n/N) - 1$. The lattice is $\mathbb{Z}^K$ for both parities. The tree creates no additional closing constraint or lattice-index Gaussian factor.

6.1 The absolute action and the original window

For $E_0 = \prod_{a=1}^{D} e_{n_a}$ with all $n_a \le m$, the complete ordinary-circle trace is

$$\phi_0(L_N E_0) = N \sum_{r=0}^{N-1} E_0(c_r), \qquad N(\pi/4)^D E_0(-2) \le \phi_0(L_N E_0) \le N^2 E_0(-2). \tag{6.4}$$

The root sum is the complete trace formula of [1, Section 6]. Its terms are nonnegative. At odd levels the node -2 proves the lower bound with constant one. At even levels the nearest node satisfies

$$\frac{e_n(-2\cos(\pi/N))}{e_n(-2)} = \prod_{a=0}^{n-1} \frac{\cos(a\pi/N)\cos((a+1)\pi/N)}{\cos^2((a+1/2)\pi/N)} \ge \frac{N}{2}\tan\frac{\pi}{2N} \ge \frac{\pi}{4}.$$

Each ratio in the product is at most one, and its endpoint value is the displayed telescoping sine product. This proves both parity bounds without a simple-node assumption at even levels.

The pure leaf has exact coefficients $[e_r]h_N = N^2 S_{2r}/S_r^4$ for $r \le m$, and zero otherwise: the root sum picks the zero Laurent coefficient because $r < N$; high e_r vanish at every complete node. Expanding E_0 by (6.1) is therefore the same positive expansion of $\phi_0(L_N E_0)$ obtained by adjoining that leaf. Uniform factorial rates and polynomial state counting in (6.4) identify its conditional maximum exactly as $\sum_a g(n_a/N)$. At the original strict state the eliminated variables have negative definite Hessian. Analytic elimination consequently identifies this function and all its derivatives near the projected state, including its additive constant.

The grouped zero-angle action is thus

$$\mathcal{A}(\mathbf{y},0)=\sum_{\text{bare gates}} m(x,y,z)+\sum_{v\ \text{circle}}\sum_{e\ni v} g(x_e). \tag{6.5}$$

Its unique maximum is the original F_0. Its Hessian in the primitive coordinates is negative definite: each circle incidence is covered by $g''<0$, and every other variable by a strict gate whose Hessian was proved negative in (4.3). The original complete positive expansion gives a strict compact complement gap, including its closed faces.

Choose a small rectangular box $\mathcal{W}$ about this maximum. All gate factorial arguments are bulk, every circle-incident label lies in a compact subinterval of $(0,1/2)$, and the incident sum at each bare vertex is greater than one by a fixed amount. The last assertion also follows by summing $x+y-z>0$ over its gates. Lemma 4.2 makes all bare physical labels strictly low except for the internal edge of the entire two-trivalent-hub component, which is half. Keep that edge in a compact subinterval of $(0,1)$; both cubic traces are exactly unchanged there. Hidden unturned indices need not be low. Every integer point of the entire box is supported for all sufficiently large N.

Before continuing any angle, discard the component errors of Theorem 5.1. For the clean contraction the actual fixed-power circle trace satisfies

$$|\phi_\sigma(R_\sigma^n E_0)|\le C_{n,R,D}N\phi_0(L_N E_0). \tag{6.6}$$

Indeed at a shifted root the angular imaginary part is $O(1/N)$. Positive Laurent coefficients bound $|E_0|$ there by $CE_0(-2)$; the fixed-power chirp has bounded modulus. The exact root formula gives $CN^2E_0(-2)$ on a larger parameter circle avoiding $\sin 2\pi\sigma=0$. Divide by (6.4) and use the maximum principle for the whole holomorphic trace to include collisions. Fixed nonreal compacts use the root formula directly. Constant orientation types retain their recorded scalar. Thus the clean resolved sum outside $\mathcal{W}$ has the original positive compact gap. This bounds arbitrary subsets of the resolved positive sum, rather than inferring an uncontracted estimate from a scalar norm. All these source losses are fixed before the powers and the meridian compact.

The retained exact packet is

$$B_N(\boldsymbol{\tau},s)=\sum_{\mathbf{k}/N\in\mathcal{W}}\left(\prod_g \mu_{N,g}\right)\left(\prod_{v\ \text{circle}}\prod_{e\ni v} e_{n_e}(-2\cos\tau_v)\right)\left(\prod_e \gamma_{e,N}(s)\frac{H_N(s)}{W_{n_e}(s)}\right). \tag{6.7}$$

It is separately even and periodic in every angle. Crossing chirps and sine trace factors are not part of this definition.

6.2 A holomorphic interpolation of this finite sum

Use the fixed canonical G_N of [5, Lemma 2.1], which interpolates S_k, satisfies its sine shift and has uniform bulk expansions with all fixed derivatives. For fixed c,

$$G_N(Nx+c)=\sqrt{2N\sin\pi x}\,e^{NJ(x)}(2\sin\pi x)^c(1+O(N^{-1})). \tag{6.8}$$

Replace every factorial in (6.1) by this function at its exact affine argument. For a circle incidence set $u=N\tau/(2\pi)$ and

$$\mathcal{E}_N(x,\tau)=\frac{G_N(d/2+u)G_N(d/2-u)}{G_N(d/2-Nx+u)G_N(d/2-Nx-u)G_N(Nx)^2}. \tag{6.9}$$

The shift identity proves equality with $e_n(-2\cos\tau)$ at $x = n/N$. Keeping $d/2 = N/2 - 1/2$ in (6.8) gives

$$\mathcal{E}_N(x,\tau) = \frac{e^{Nb(x,\tau)}}{2N\sin\pi x}(1 + O(N^{-1})), \tag{6.10}$$

$$b(x,\tau) = \int_0^x \log(2\cos 2\pi t + 2\cos\tau)\,dt - 2J(x). \tag{6.11}$$

The branch continues the positive real integrand at $\tau = 0$; $b(x,0) = g(x)$. An index $N\ell + c$ contributes the amplitude $e^{cb_x(\ell,\tau)}$. All these fixed offsets are retained.

The edge interpolation is exactly (1.9): the sine shift and $G_N(s)G_N(-s) = 1/H_N(s)$ give $\mathcal{I}_N(n/N,s) = H_N(s)/W_n(s)$. On the fixed bulk tube,

$$\mathcal{I}_N(x,s) = 1 - \frac{\pi s^2}{N}\cot\pi x + O(N^{-2}). \tag{6.12}$$

In fact, if $L_N = \log G_N$, then $2L_N(Nx) - L_N(Nx-s) - L_N(Nx+s) = -s^2 L_N''(Nx) + O(N^{-3})$ and $L_N''(Nx) = \pi\cot(\pi x)/N + O(N^{-2})$.

Let $\mathcal{A}(\mathbf{y},\boldsymbol{\tau})$ be the sum of the gate actions and these b's. Dividing the exact interpolated summand by $\Gamma_{e,N} = \prod_e \gamma_{e,N}$ gives

$$N^{-D_c - 3G/2}\beta_N(\mathbf{y},\boldsymbol{\tau})e^{N\mathcal{A}(\mathbf{y},\boldsymbol{\tau})}\prod_e \mathcal{I}_N(n_e/N,s), \tag{6.13}$$

where $\beta_N = \beta + O(N^{-1})$ is holomorphic and $\beta(\mathbf{y}^*,0) > 0$. The implicit function theorem gives a unique local wire saddle $y_*(\boldsymbol{\tau})$ and invertible $Q = -\mathcal{A}_{yy}(y_*,\boldsymbol{\tau})$, with $Q(0) > 0$.

Here is the finite-sum justification of the Gaussian. Enclose each retained integer interval by a rectangle with half-grid vertical sides and horizontal heights $\pm h$. The product residue formula with kernels $\pi N\cot(\pi N z)$ is exactly the lattice sum. On a vertical side $|\cot| \le 1$, while on the horizontal sides

$$\cot(\pi N(x \pm \mathrm{i}h)) = \mp\mathrm{i} + O(e^{-2\pi Nh}).$$

At zero angles the phase gain under the imaginary shift is at most Ch^2 and the real boundary has a fixed loss δ. Choose h with $Ch^2 < \min(\delta/4, \pi h)$, then shrink the angular polydisk. Every vertical side and every nonconstant cotangent mode is exponentially smaller. The constant terms give N^K times the real-box integral; its endpoint displacements of $O(1/N)$ have the same boundary loss. Translate that box to its complex saddle. Its connecting sides retain the loss, and its real tangent plane has negative real Hessian. Gaussian expansion gives

$$\Psi(\boldsymbol{\tau}) = \mathcal{A}(y_*,\boldsymbol{\tau}), \qquad a(\boldsymbol{\tau}) = \frac{(2\pi)^{K/2}\beta(y_*,\boldsymbol{\tau})}{\sqrt{\det Q}}.$$

The square root starts at the positive real Gaussian and never vanishes on a smaller polydisk. Equation (6.3) gives the claimed power of N. Odd Gaussian terms integrate to zero, so the relative remainder is $O(1/N)$, uniformly holomorphic. Separate evenness and uniqueness imply $\mathcal{A}_{y\tau_v}(\mathbf{y},0) = 0$. Direct differentiation gives $b_{\tau\tau}(x,0) = -\tan(\pi x)/(2\pi)$, proving the diagonal angular Hessian in Theorem 1.1.

6.3 The whole real angular cube

For n/N in a fixed compact subset of $(0,1/2)$, the exact product gives the following strengthening of [4, Lemma 7.2]:

$$|e_n(-2\cos(t + \mathrm{i}y))| \le N^A e^{Ng(n/N) - Nct^2 + NCy^2}, \qquad |t| \le \pi.$$

Indeed near zero the logarithm of $e_n(-2\cos w)/e_n(-2)$ is even, has second derivative $-2\sum_{j<n}(c_j+2)^{-1} \le -cN$, and all fixed higher derivatives are $O(N)$ on one uniform zero-free disk. Taylor's formula gives $-cNt^2 + CNy^2$ there. On the remaining compact real interval the imported lemma gives a strict real gap, including sampled numerator zeros, by truncated logarithmic Riemann upper sums. Shrink the complex height and reduce c so a fixed part of that gap remains; it absorbs the uniform $e_N = o(1)$ error in the imported estimate. Finally $e_n(-2) = \exp(Ng(n/N) + O(\log N))$ supplies the polynomial prefactor. Thus no polynomial error bound at arbitrary numerator collisions is assumed. All bare factors in (6.7) are nonnegative and the edge ratios are bounded. The positive zero-angle sum has rate F_0 and polynomial prefactor by the Gaussian just proved. Every circle has an incident edge, so summing the bound proves (1.7) with a positive loss in every angle. Endpoints translated by $O(1/N)$ retain it by periodicity: the squared distance to $2\pi\mathbb{Z}$ changes by $O(1/N)$. Fixed derivatives cost powers of N by Cauchy estimates on radius-$1/N$ circles. This completes the proof of Theorem 1.1, including its actual finite source error before any crossing contour is moved.

7 All charges of the original crossing traces

The powers n_v are fixed signed integers. In the primitive filling convention they are $n_v = \epsilon_v - 2p_v$; odd powers are included. This section proves Theorem 1.2 analytically. It does not identify a geometric action.

7.1 The full finite functional and its currents

For an even polynomial $f(\tau) = f^\sharp(-2\cos\tau)$ define $\mathcal{D}_{n,\sigma,N} f = \phi_\sigma(R_\sigma^n f^\sharp)$. The full crossing data in [1, Proposition 6.1 and equation (6.4)] give, with $\eta = (-1)^d$,

$$\mathcal{D}_{n,\sigma,N} f = \frac{\eta N C_{N,n}(\sigma)}{\sin 2\pi\sigma} \sum_{r=0}^{N-1} \sin\tau_r \, f(\tau_r) e^{\mathrm{i}nN\tau_r^2/(4\pi)}, \tag{7.1}$$

$$\tau_r = \frac{2\pi}{N}(d/2 + \sigma - r), \qquad C_{N,n}(\sigma) = q^{n(d^2/4-\sigma^2/2)}. \tag{7.2}$$

This is first written off the collisions. Its removable limit is the full Hermite functional, not just the node values. In particular inverse powers use the same formula.

Introduce independent formal grid parameters σ_v temporarily; the actual assignments are $0, s, -s$. If ζ_N is the original closing and framing scalar and $\rho_{v,N}$ the reference scalar for its orientation type, set $\widehat{\zeta}_N = \zeta_N \prod_v \rho_{v,N}$. The exact source reduction already proved has the form

$$F_N(q^s) = \widehat{\zeta}_N(s) H_N(s) \left(\prod_v \mathcal{D}^{(v)}_{n_v,\sigma_v,N} \right) B_N(\boldsymbol{\tau}, s) + O(N^A e^{N(F_0-\eta_0)}). \tag{7.3}$$

Each factor acts in its own variable; this is tensor evaluation, not commutation of physical crossing operators. Edge scalars are already in B_N and are not counted in $\widehat{\zeta}_N$.

Choose $R < R' < 1$, $R' \neq 1/2$, and initially take $|\sigma_v| = R'$. For one coordinate use the exact pole kernel

$$K_\sigma(\tau) = \frac{\mathrm{i}N e^{\mathrm{i}N\tau}}{e^{\mathrm{i}N\tau} - \eta e^{2\pi\mathrm{i}\sigma}}.$$

It has residue one at precisely the shifted grid points. Integrate it times $\sin\tau f(\tau) e^{\mathrm{i}nN\tau^2/(4\pi)}$ around a rectangle translated by $2\pi\sigma/N$ enclosing those N poles. On its vertical sides

$$K_\sigma(\pm\pi + \mathrm{i}v + 2\pi\sigma/N) = \frac{\mathrm{i}N e^{-Nv}}{1 + e^{-Nv}}.$$

On horizontal sides expand the denominator geometrically. After reflecting the upper side by $\tau \mapsto -\tau$, the odd sine factor pairs the positive and negative frequencies. Their parameter multiplier is

$$\frac{e^{2\pi \mathrm{i}\ell\sigma} - e^{-2\pi \mathrm{i}\ell\sigma}}{e^{2\pi \mathrm{i}\sigma} - e^{-2\pi \mathrm{i}\sigma}} = U_{\ell-1}(\cos 2\pi\sigma).$$

The grid sign and the contour orientations give

$$c_{\ell,N}(\sigma) = \eta^{\ell-1} U_{\ell-1}(\cos 2\pi\sigma), \qquad c_{1,N} = 1. \tag{7.4}$$

The zero-frequency real integral vanishes by oddness. Its endpoint currents, and the connectors from the translated endpoints, remain in the contour identity. This calculation explains why replacing the repeated-node functional by values would lose the factor $c_{\ell,N}(0) = \eta^{\ell-1}\ell$.

Choose $t_0 > 0$ small and put $L_v = \lfloor t_0|n_v| \rfloor$, $y_v = A_0 t_0/|n_v|$. The finite real functional retained from this exact contour is

$$\mathcal{P}^L_{n,\sigma,N} f = -\frac{N^2 C_{N,n}(\sigma)}{\pi \mathrm{i}} \sum_{\ell=1}^{L} c_{\ell,N}(\sigma) \int_{-\pi}^{\pi} \sin\tau\, f(\tau) e^{\mathrm{i}nN\tau^2/(4\pi) - \mathrm{i}\ell N\tau}\, d\tau. \tag{7.5}$$

Here is a uniform bound for what was omitted. On the lower horizontal side the packet and chirp have real rate at most

$$F_0 - ct^2 + \frac{|n||t|y}{2\pi} - \ell y + Cy^2.$$

The maximum of the two t terms is $n^2y^2/(16\pi^2 c)$. For $\ell > L$ the resulting upper rate is

$$F_0 + t_0^2 \left(\frac{A_0^2}{16\pi^2 c} - A_0 + \frac{CA_0^2}{n^2} \right) - (\ell - L - 1)y.$$

First choose A_0 so its first positive term is at most $A_0/4$, then a lower threshold for $|n|$ so the last one is also at most $A_0/4$. The loss is at least $A_0 t_0^2/2$, independently of n. The bound $|c_{\ell,N}(\sigma)| \le \ell e^{2\pi R'(\ell-1)}$ makes the remaining geometric sum convergent for all sufficiently large N. Constants depending on fixed n do not change this rate.

At the vertical endpoints the angular loss is fixed. Their chirp cost is $O(N|n|y) = O(Nt_0)$, and the retained-charge connector cost is $O(NLy) = O(Nt_0^2)$. Decrease t_0 to preserve the loss. This also controls the zero-mode current. All spectator variables can remain real or on their physical translated grids: those grid heights are $O(1/N)$ and their fixed chirps have bounded modulus. Use the exact telescoping identity

$$\prod_v \mathcal{D}_v - \prod_v \mathcal{P}_v = \sum_j \Big(\prod_{v<j} \mathcal{P}_v\Big)(\mathcal{D}_j - \mathcal{P}_j)\Big(\prod_{v>j} \mathcal{D}_v\Big).$$

The other $\mathcal{P}$'s are finite real integrals of polynomial operator cost, and the other $\mathcal{D}$'s are finite physical grid sums. Thus every telescoping term has rate $F_0 - \gamma$ for one fixed $\gamma > 0$. No spectator is moved to a slope-dependent complex height during another coordinate's tail estimate.

The whole trace-minus-finite-approximation is holomorphic in each σ_v. Apply the maximum principle successively to extend the estimate from the parameter circles to $|\sigma_v| \le R$. This includes all normal limits at $0, \pm 1/2$. For any fixed upper-half-plane compact use the same translated contour directly; its sine denominator is bounded away from zero and its endpoint translation is still $O(1/N)$. Constant vertices can independently be brought to zero by the preceding disk argument. Fixed derivatives follow on smaller compacts.

7.2 A common critical branch for every retained charge

After truncating the real integral to a small angular box, with fixed loss, its phase is

$$T_{\boldsymbol{\ell}}(\boldsymbol{\tau}) = \Psi(\boldsymbol{\tau}) + \frac{\mathrm{i}}{4\pi}\sum_v n_v\tau_v^2 - \mathrm{i}\sum_v \ell_v\tau_v.$$

Set $\varepsilon_v = 1/n_v$, $t_v = \ell_v/n_v$. Its critical equation and a removable stationary expression are

$$\tau_v = 2\pi t_v + 2\pi\mathrm{i}\varepsilon_v\Psi_v(\boldsymbol{\tau}), \tag{7.6}$$

$$W(\mathbf{t},\boldsymbol{\varepsilon}) = \Psi(\boldsymbol{\tau}) - \mathrm{i}\pi\sum_v \varepsilon_v\Psi_v(\boldsymbol{\tau})^2. \tag{7.7}$$

The implicit function theorem gives one holomorphic branch on a fixed polydisk about $(0,0)$. The critical action is $-\mathrm{i}\pi\sum t_v^2/\varepsilon_v + W$; the first term is purely imaginary for the real signed data. Separate evenness makes W even in each t_v. At $\boldsymbol{\varepsilon} = 0$, $W(\mathbf{t},0) = \Psi(2\pi\mathbf{t})$, so $\partial_{t_v^2}\Re W(0,0) = -2\pi^2 a_v$. Shrink the polydisk until all these derivatives are at most $-\pi^2 a_v$. Integration in the squared coordinates gives

$$\Re T_{\boldsymbol{\ell}}(\boldsymbol{\tau}_{\boldsymbol{\ell}}) \le \Re T_{\mathbf{1}}(\boldsymbol{\tau}_{\mathbf{1}}) - \sum_v \frac{\pi^2 a_v}{n_v^2}(\ell_v^2 - 1). \tag{7.8}$$

Choose t_0 inside this polydisk and then increase the single lower threshold for all $|n_v|$. This covers every retained charge without a restriction on their slope ratios.

Write $\Psi_v = \tau_v Q_v(\tau_1^2,\dots,\tau_H^2)$, with $\Re Q_v < 0$ in that neighborhood. Then

$$\tau_{\boldsymbol{\ell},v} = \frac{2\pi t_v}{1 - 2\pi\mathrm{i}\varepsilon_v Q_v}, \qquad \Im\tau_{\boldsymbol{\ell},v} = \frac{4\pi^2\ell_v\Re Q_v}{n_v^2|1 - 2\pi\mathrm{i}\varepsilon_v Q_v|^2} < 0. \tag{7.9}$$

Every angle and every sine in the amplitude is nonzero. Translate the small real box to this critical point. The real part of its Hessian on real tangent vectors is that of Ψ'', since the chirp Hessian is purely imaginary, and stays negative definite. The connectors preserve the face loss: their heights are at most $C|t_v|/|n_v|$, their chirp cost on a face of size ρ is at most $CN\rho|t_v|$, and their Fourier factors decrease under the downward displacement. Choose t_0 small compared with the fixed face loss before increasing the slope threshold.

Use the holomorphic packet expansion on this contour, not a real pointwise remainder. Gaussian integration and (6.3) give for the entire retained contraction

$$\left(\prod_v \mathcal{D}_v\right) B_N = \left(\prod_v C_{N,n_v}(\sigma_v)\right)\Gamma_{e,N}(s)N^{3/2}D_{\mathbf{n}}e^{NT_{\mathbf{1}}(\boldsymbol{\tau}_{\mathbf{1}})}(1 + O(N^{-1})), \tag{7.10}$$

where

$$D_{\mathbf{n}} = \left(-\frac{1}{\pi\mathrm{i}}\right)^H (2\pi)^{H/2}\frac{a(\boldsymbol{\tau}_{\mathbf{1}})\prod_v \sin\tau_{\mathbf{1},v}}{\sqrt{\det[-T_{\mathbf{1}}''(\boldsymbol{\tau}_{\mathbf{1}})]}} \neq 0.$$

The branch is fixed by the translated real Gaussian. There are only finitely many charges for fixed powers. Equation (7.8) suppresses every one except $\mathbf{1}$, whose coefficient is exactly one by (7.4). The reflected signs have already been paired, leaving no equal-action terms with an unexamined relative sign. The wire Hessian and this angular Schur complement are invertible, hence so is the full joint Hessian.

The first action tends to F_0 as $\max_v |1/n_v| \to 0$. Increase the common threshold to put its real part above F_0 minus half each fixed source, contour-tail and off-window loss. Those additive errors are then negligible. The charge-dependent losses were instead compared

to this same filled action in (7.8); a smallest local loss is never compared with a total complete-to-filled drop.

Finally restrict σ_v to the actual types. The finite scalar ledger [1, Proposition 8.1], which retains every typed turn, monodromy, crossing carrier and framing correction, says

$$\frac{\widehat{\zeta}_N(s)}{\widehat{\zeta}_N(0)}\Gamma_{e,N}(s)\prod_v q^{-n_v\sigma_v(s)^2/2}=q^{a(D)s^2/2},\qquad a(D)=-w(D)-\sum_v n_v.$$

There is still exactly one main-cut factor H_N in (7.3). This proves (1.8). Removing these exact scalars leaves only the product of (1.9) on the original physical edges in the joint integral. The zero-meridian amplitude and all its offsets are common, so no first-order scalar is hidden in that assertion. Every choice of an exponential margin or a slope threshold preceded the choice of a meridian compact; only bounded factors and starting levels depend on that compact.

References

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